GIT-207

Common infections of mouth, salivary glands, esophagus and stomach

Microbiology & Immunology Department

By

Dr/ Shereen Mohamed Mohamed

Associate professor of Medical Microbiology and Immunology

Objectives:

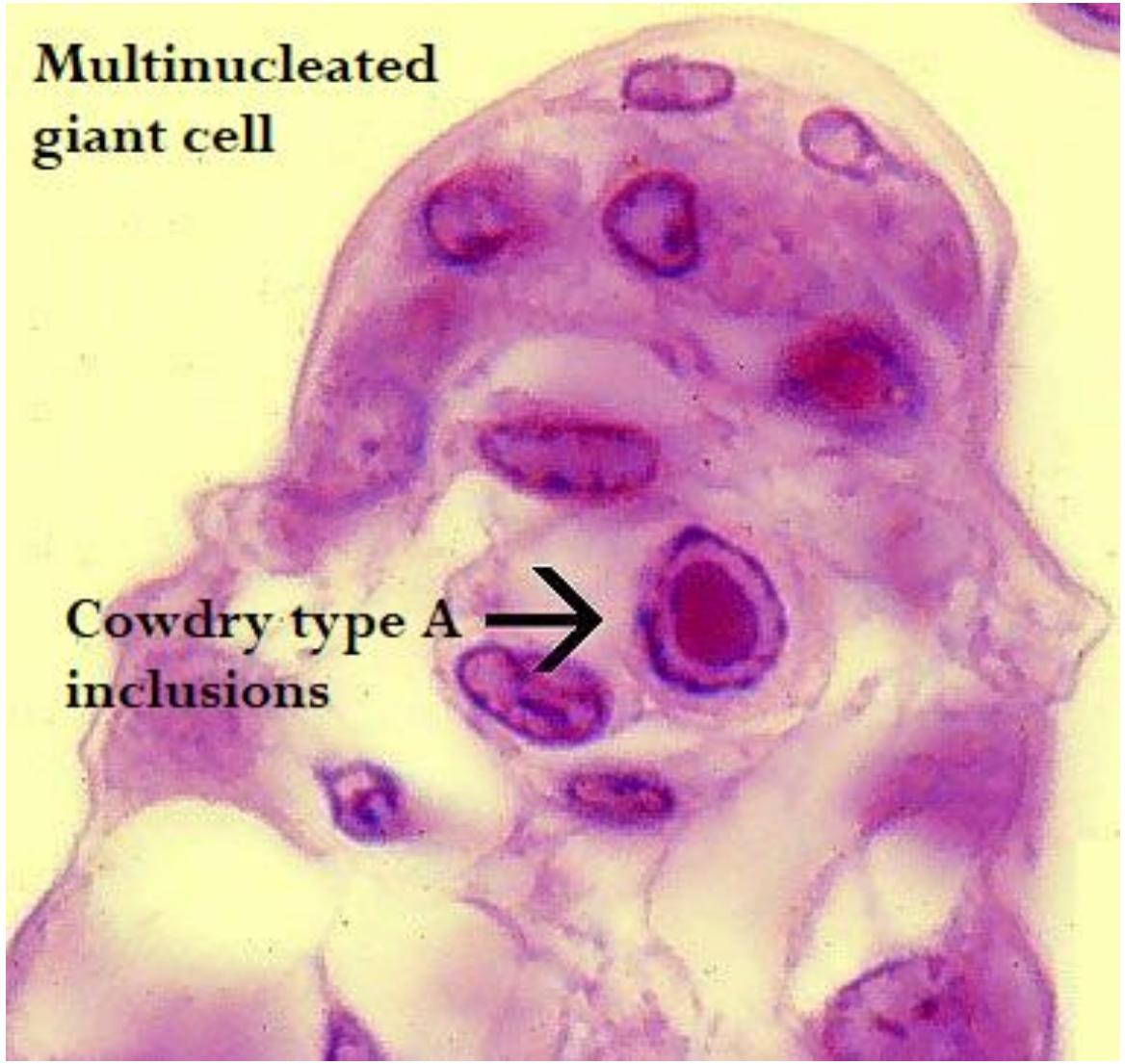
By the end of this lecture, the student will be able to:

- Recall the causes of common infections of mouth, their distinguishing features and laboratory diagnosis.
- Define the causative agent, mode of transmission, laboratory diagnosis and method of prevention of mumps.
- Explain the causes of common infections of esophagus, their clinical presentation and diagnosis.
- Describe the features of H. pylori, mode of transmission, pathogenesis and diagnosis.

Common Infections of the Mouth

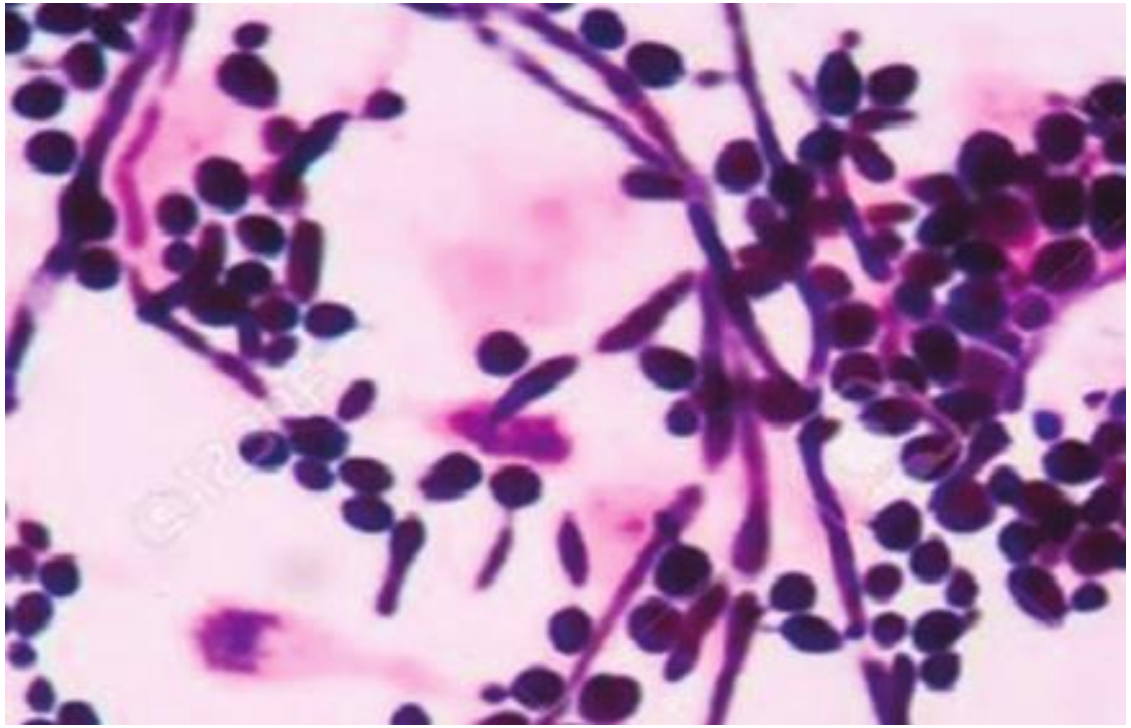
Herpetic stomatitis- gingivostomatitis (Cold sores)

Pathogen	HSV-1
Clinically	Vesicular lesions on oral mucosa
Diagnosis	Tzanck smear to show multinucleated giant cells and Cowdry type A intranuclear inclusions
Prognosis	The virus remains latent in the trigeminal ganglion with repeated reactivation (stress triggers reactivation)


Multinucleated
giant cell
Cowdry type A
inclusions

Oropharyngeal candidiasis (Thrush)

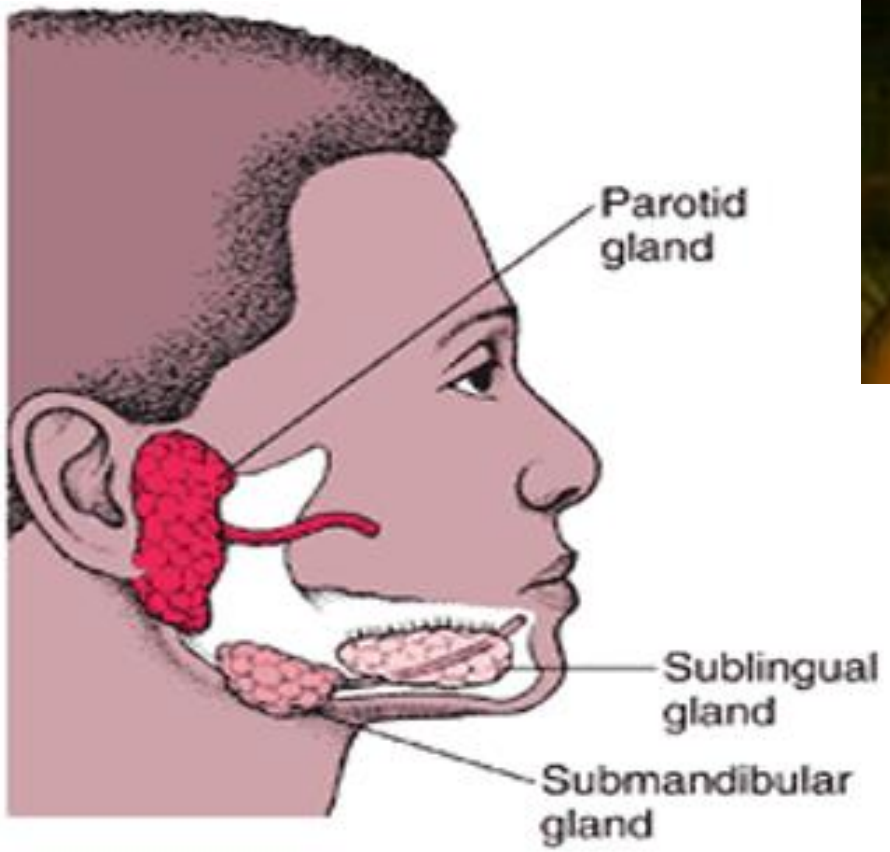
Pathogen	*Candida albicans*
Clinically	Painful mouth lesions coated with thick white discharge with red base under the coat.
Diagnosis	Oval, budding yeast cells forming hyphae and pseudohyphae are detected in 10% KOH wet mount or Gram-stained smear prepared from the discharge
Predisposing factors	Occur in immunocompromised hosts e.g. premature infants, patients on antibiotics or steroids, diabetes, AIDS.



Common diseases of the salivary glands

Mumps

Pathogen	Mumps virus (Paramyxoviridae family, enveloped ⊖ SS RNA virus)
Transmission	Person to person by respiratory droplets
Clinically	▪ May be asymptomatic OR ▪ Presented as bilateral parotitis with fever, headache, and malaise
Complications	▪ Pancreatitis ▪ Orchitis (inflammation of testes) causing sterility in males Meningoencephalitis
Diagnosis	▪ Clinical picture Serology: e.g. ELISA
Prevention	▪ MMR (Measles/Mumps/Rubella) vaccine (live attenuated vaccine)


Parotid gland
Sublingual gland
Submandibular gland

Common Infections of The Esophagus

Usually occur due to extension of oropharyngeal infections usually in immunocompromised patients.

Commonly caused by:

- *Candida albicans*
- Herpes simplex
- Cytomegalovirus

Clinically presented as painful swallowing

Diagnosis by: Endoscopy and biopsy

Common infections of the stomach

Helicobacter pylori

Features

- Gram-negative motile helical bacilli
- Microaerophilic
- Triple ⊕:
  - Urease-positive
  - Oxidase-positive
  - Catalase-positive

Transmission: Direct from person to person (Fecal-oral/ Oral-oral)

Pathogenesis: *H. Pylori* produces

- Urease enzyme: converts urea to ammonium that neutralizes stomach acid, allowing bacterial survival in stomach acid.
- Mucinase enzyme: aids in penetration of mucous layer of the stomach.

Diseases

- Chronic gastritis and Peptic ulcers
- Several forms of stomach cancer (e.g. adenocarcinoma)
- Now *H. pylori* is classed by WHO as type I carcinogen

Clinically:

* Epigastric pain
* Dyspepsia
* Nausea & vomiting
* Hematemesis

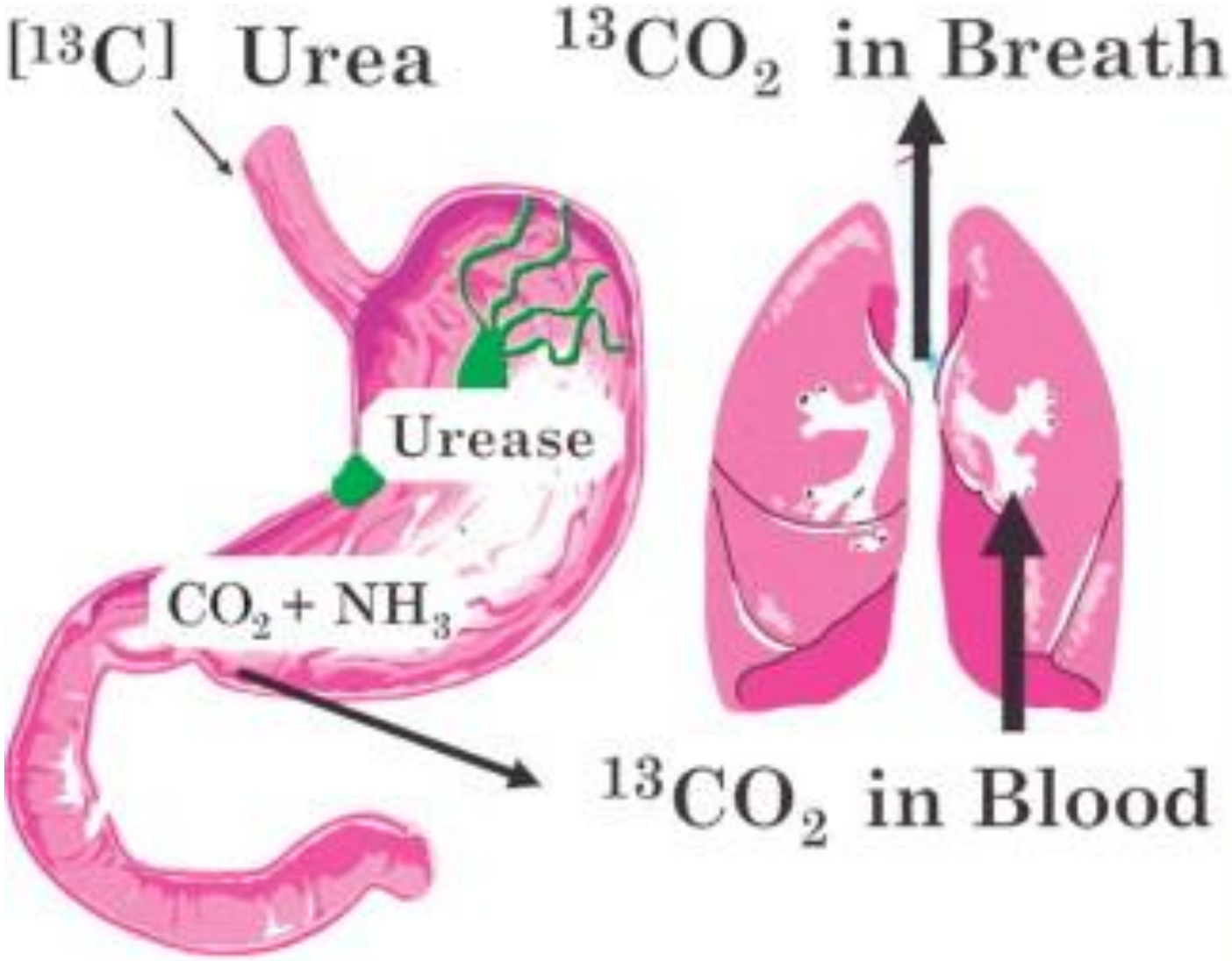
Diagnosis:

A. Invasive methods: Gastroscopy and biopsies → gold standard for diagnosis

B. Non-invasive methods:

1. *H. pylori* stool antigen test: detects *H. pylori* antigens in a stool sample.
2. Urea breath test: detection of $^{13}Co_2$ in breath samples

- Urea containing labeled carbon isotope (^{13}C) is swallowed.
- If the individual is infected with *H. pylori*, ammonia is produced and $^{13}CO_2$ is exhaled)



Breath Collection

ANY
QUESTIONS?